

ENHANCING STUDENT LEARNING THROUGH AUTOMATED ASSISTANCE ...

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INTRODUCTION

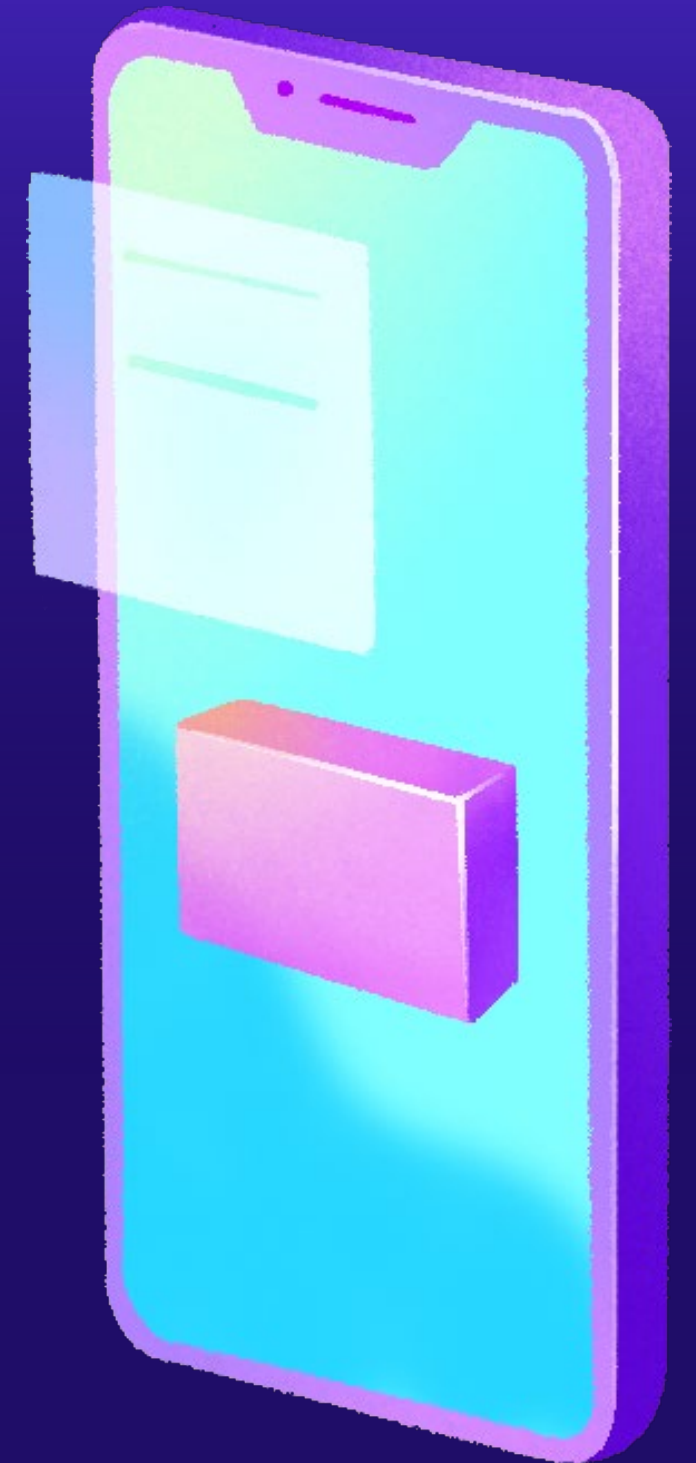
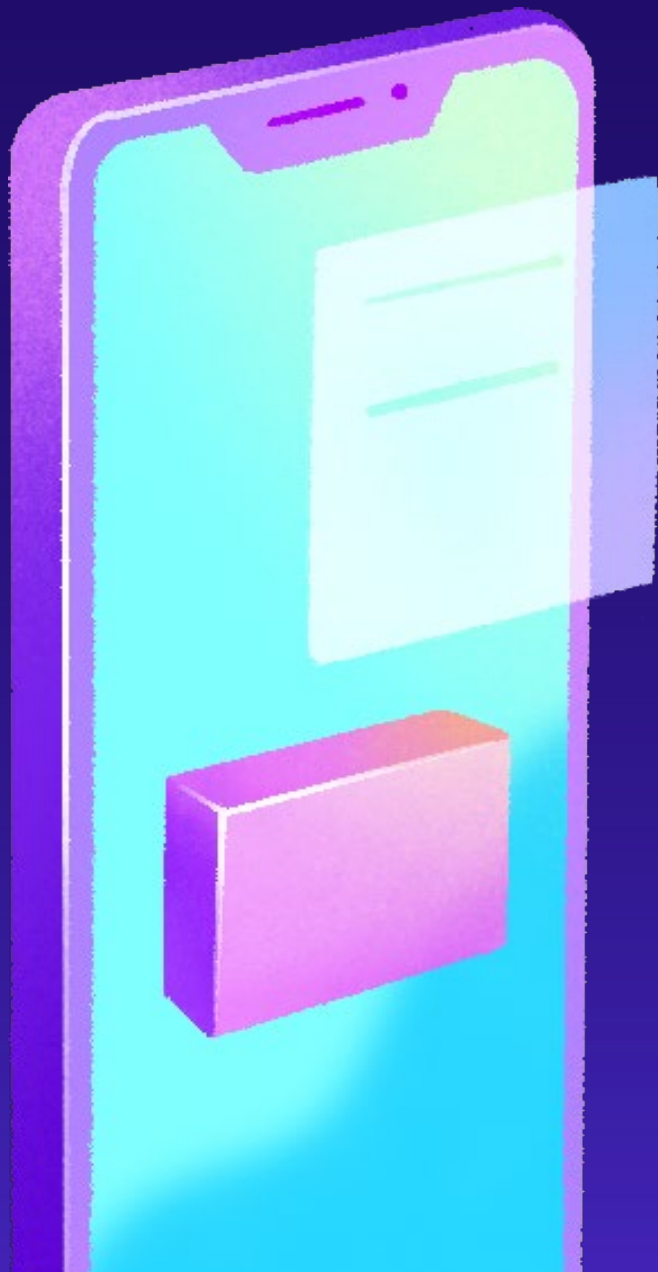
This project aims to create an automated Q&A system to assist students in finding answers to questions from course materials or textbooks, enhancing learning by providing instant, accurate responses.



QUESTED

PROBLEM STATEMENT

Students struggle to efficiently find answers in extensive study materials, causing interruptions in their learning. This project seeks to resolve this by implementing a question-answering AI model.



LITERATURE REVIEW

Existing Models : BERT, GPT, RoBERTa

These models have shown remarkable effectiveness in understanding and generating human -like text.

General Q&A Systems : ChatGPT, IBM Watson

provide general -purpose assistance, they lack the tailored focus needed for academic content.

This project aims to address these gaps by fine -tuning a Q&A model specifically for educational contexts, enhancing accuracy and relevance for student users.



OBJECTIVES

1. AI-Powered Q&A System

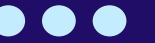
Develop a robust system tailored for student assistance

2. NLP Integration

Utilize advanced NLP for accurate query interpretation and response

3. User-Friendly Interface

Create an intuitive interface that integrates with educational resources



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PROPOSED SOLUTION

AI Approach

This system will leverage a fine-tuned transformer model, such as T5 or GPT-2, specifically trained on educational content. Transfer learning will be utilized to adapt a pre-trained model to handle specific types of queries relevant to various subjects.

Data Requirements

- Datasets containing textbooks, course materials, and question-answer pairs.
- Data preprocessing will include Academic tokenization, filtering, and structuring of educational texts to optimize model performance.



PROJECT DEPENDENCIES & TECHNOLOGIES

Key Functionality:

- Upload & process files (PDFs images)
- Extract text using OCR
- Store & retrieve data from databases
- NLP processing for text analysis

Languages & Tools:

- Python/ HTML/ CSS
- Node.js
- XAMPP
- VsCode

Backend Framework:

- Express.js

Database:

- MySQL (mysql2)

File Handling:

- Multer (file uploads)
- PDF-lib & pdf-parse (PDF processing)

OCR & NLP:

- Tesseract.js (text extraction)
- Hugging Face Transformers (NLP models)

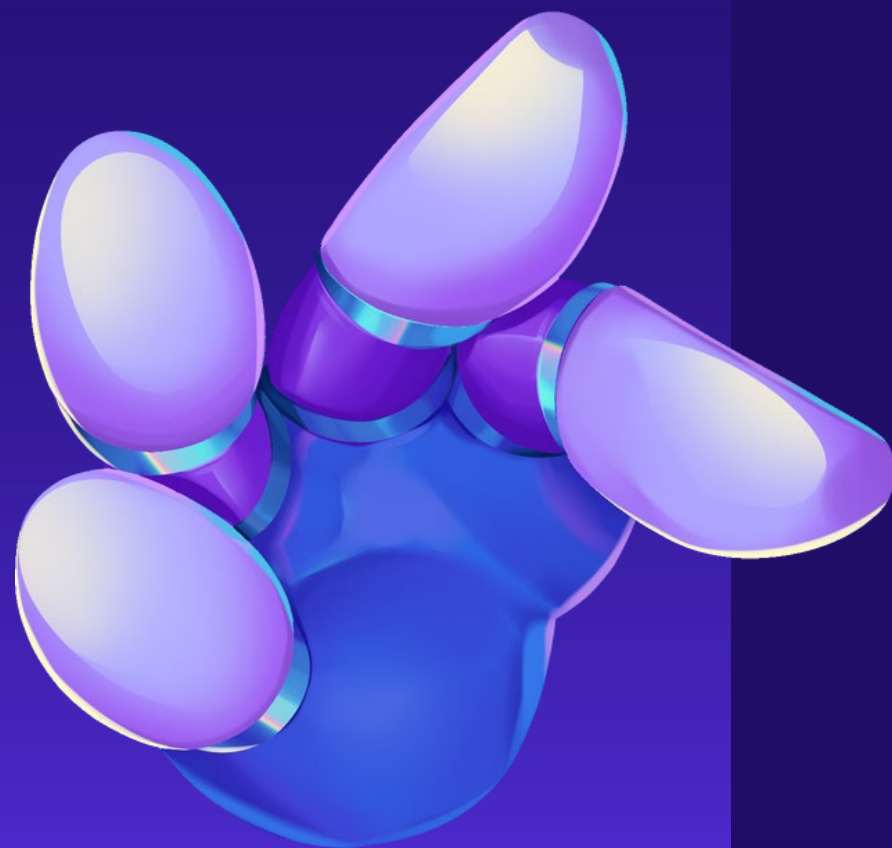
HTTP Requests:

- Axios (API communication)



EVALUATION

The effectiveness of the Q&A system will be evaluated using several key metrics. Accuracy will measure the percentage of correct answers out of total queries. Precision and recall will assess the relevancy and completeness of retrieved answers. The F1 score will provide a balanced evaluation of the system's overall effectiveness



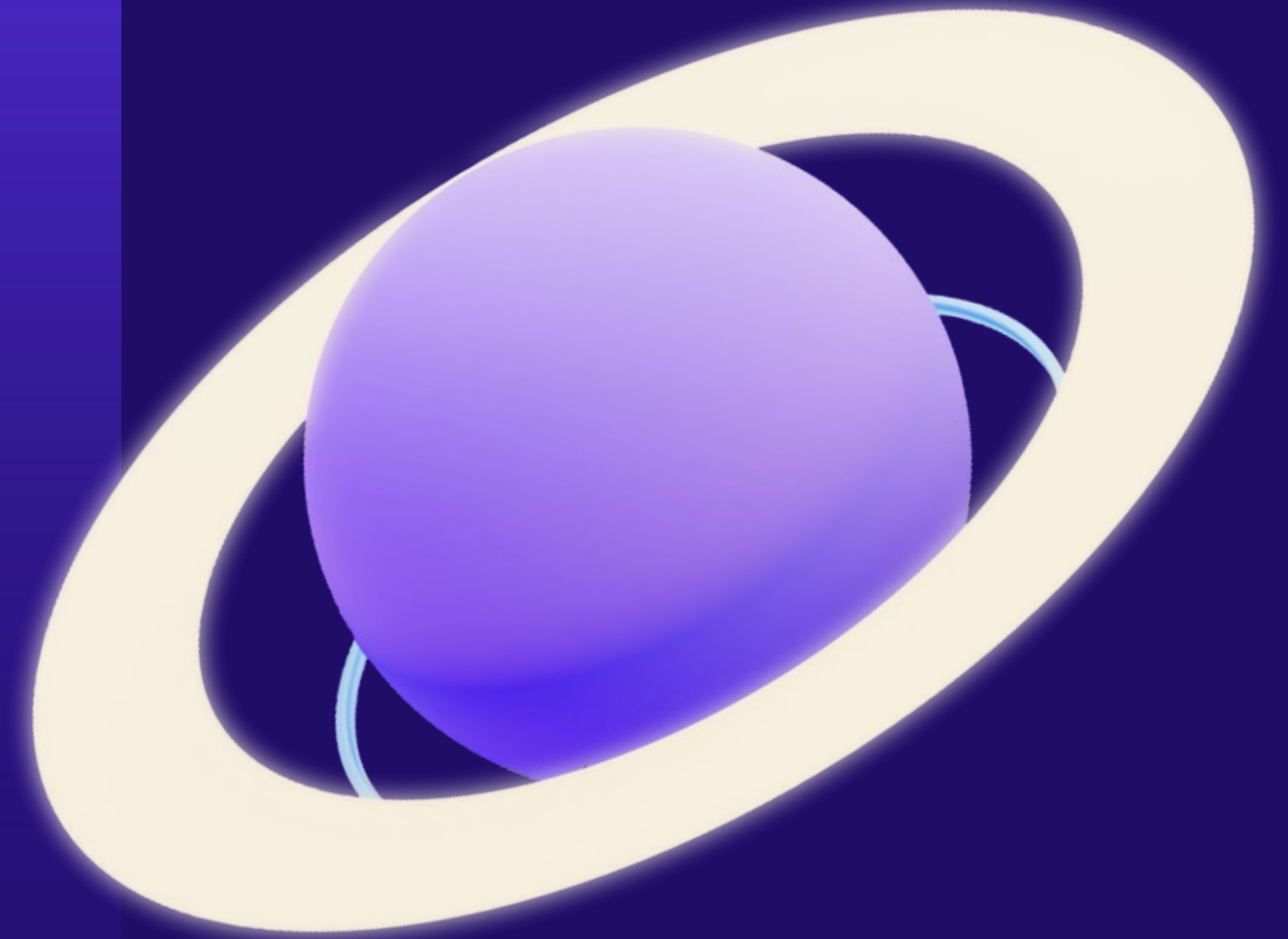
	T5	GPT
Total Samples	5928	6798
Best Exact Match	64.22%	74.32%
Best F1 Score	73.11%	74.23%



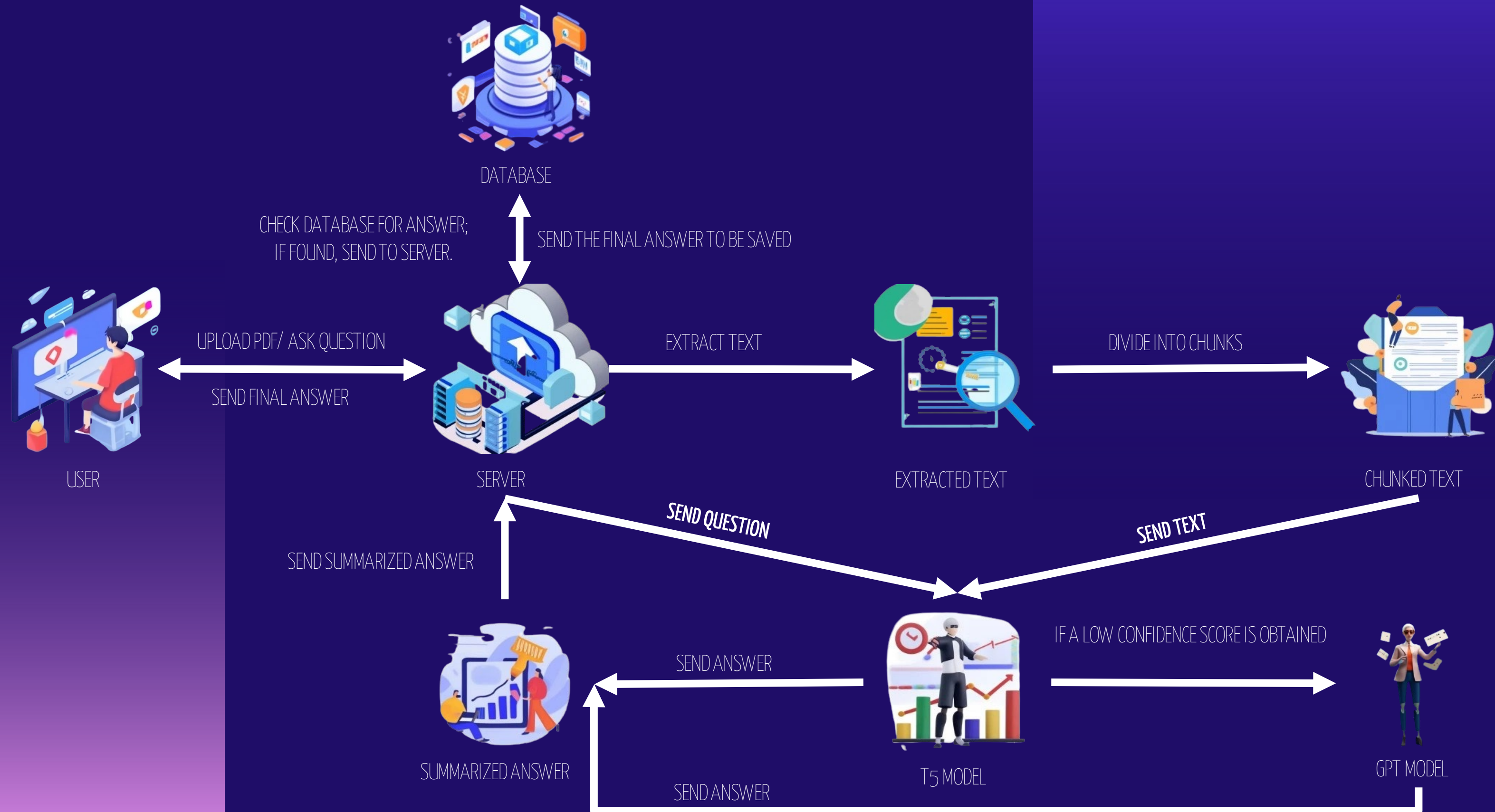
EXPECTED OUTCOMES

The Q&A system is expected to significantly reduce students' search time for information, allowing for more efficient and focused study sessions .

- **Time Efficiency:** Reduced search time for information
- **Enhanced Learning:** Improved understanding and retention
- **Academic Success:** Better overall performance and engagement



ARCHITECTURAL DIAGRAM



DEMONSTRATION



CONCLUSION

This project aims to develop an AI-powered Q&A system for instant, accurate answers from course materials, enhancing self-paced learning. By leveraging NLP, it provides students with a user-friendly tool for deeper understanding and retention, supporting personalized, efficient learning experiences.



Chat with your PDF using AI

Explore knowledge from your PDFs instantly with our easy-to-use chat interface.

Start Chatting



Q & A

THANK YOU!

